SOIL EROSION AND CONSERVATION · NOTE CM03

Terraces — Types, Spacing, Grades, Cross-Section, Planning, Construction and Maintenance

Course topics: SWCE4.5 L10, SWCE4.5 L11, SWCE-501 T4

Mechanical measures on arable land

Conservation measures on farmland are biological or engineering (mechanical); mechanical measures are used when biological ones alone are not enough. Barriers across the slope lengthen the time of concentration (helping infiltration), turn a long slope into a series of short ones (keeping runoff velocity below critical) and protect land from excess runoff. The commonest are terraces and bunds.

Terracing

- A terrace is an earth embankment or ridge across a slope to control runoff and minimise erosion. It shortens the hillside slope, reduces sheet and rill erosion and prevents gullies.
- Soil loss ∝ (slope length)^0.5, so doubling the slope length raises erosion ≈ 1.4 times; terraces shorten the length. About 70% of splash-eroded soil moves downhill, and terraces trap it in their channels.
- **Not justified** where agronomic measures (contour tillage, rotation, strip cropping) suffice, rainfall intensity is low, the soil absorbs rain quickly, or slopes are gentle. **Needed** on highly erodible soils, long slopes and under intense rain — always supplemented by good cropping.

Type	Features
Bench terrace	Reduces the land slope: a shelf with a level or nearly level top and a steep or vertical downhill face, built on the contour — for mountainous regions
Broad-base (ridge-type) terrace	Removes or retains water on sloping land: ridge 25–50 cm high and 5–9 m wide with gently sloping sides and a dish-shaped channel upslope; level or graded toward one or both ends
Graded terrace	Constant or variable grade to lead excess water safely to an outlet — interception and diversion type
Level terrace	On the true contour, no channel slope — for permeable soils where moisture conservation matters and outlets are impractical — absorption type
Channel type	Where controlled removal of water matters; built mainly from one side; key feature an excavated channel
Ridge type	Where holding water is the main aim; the ridge holds water over a wider area; usually built from both sides
Narrow-base terrace	Base 1.2–2.5 m; the ridge cannot be cultivated (the early type)
Broad-base (Mangum) terrace	Ridge widened so it can be cultivated — introduced in 1885 by Priestley H. Mangum

Terrace design

Design considers spacing and location, channel capacity, a cross-section that can be farmed, runoff removal at non-erosive velocity, and soil, cropping, soil management and climate.

Spacing

- Expressed as the **vertical interval (V.I.)** — the vertical distance between corresponding points of adjacent terraces — or the **horizontal interval (H.I.)**. V.I. is preferred for layout and construction.

$$\text{H.I.} = \text{V.I.} / \text{land slope (m/m)}$$

$$\text{Graded terraces: V.I.} = K \cdot S + Y$$

V.I. in m; S = average land slope (%); K = constant for geographical location (0.12–0.24); Y = constant for soil erodibility and cover

- **ASAE (1989) values of Y:** 0.3 (highly erodible soil, no residue), 0.6, 0.9, 1.2 (erosion-resistant soil with conservation tillage).
- Spacing may vary ±25%. Terraces are not recommended on slopes > 10–12%.

Field slope (%)	V.I. (m)	H.I. (m, as printed)	H.I. recomputed = V.I. ÷ slope
1	0.75	75	75
2	0.90	45	45
3	1.05	35	35
4	1.20	30	30
5	1.35	27	27
6	1.50	25	25
7	1.65	23.7	23.6
8	1.80	22.5	22.5
9	1.95	21.6	21.7
10	2.10	21.0	21.0
11	2.25	20.4	20.5
12	2.40	20.1	20.0
13	2.55	19.8	19.6
14	2.70	19.2	19.3
15	2.85	18.9	19.0

Table 14.6 — recommended spacing for graded terraces. Its V.I. values follow V.I. = 0.15S + 0.6 (K = 0.15, Y = 0.6). The printed H.I. values carry small rounding slips, shown in the last column.

Grade and length

- **Uniform graded terrace:** 0.1% to 0.4%, steeper for impervious soils and long terraces. **Variable graded terrace:** 0 to 0.5%, so capacity increases toward the outlet.
- **Length:** at most 400–500 m in one direction; shorter is better, to avoid erosive velocities and large sections. With outlets at both ends a terrace can drain both ways. Level terraces have no maximum

length, but earth check dams every 120–150 m are recommended.

Terrace length (m)	Lower ¼	Second ¼	Third ¼	Upper ¼
30–120	0.3	0.3	0.2	0.2
150–240	0.4	0.3	0.2	0.2
270–360	0.5	0.4	0.3	0.2
> 400	0.5	0.4	0.3	0.2

Table 14.7 — channel grades (%) for variable graded terraces, by quarter of length from the outlet end.

Cross-section and capacity

- Needs adequate capacity, slopes flat enough for machines, and economical construction. Channel-type terraces get capacity from the excavated graded waterway; ridge-type from the water held by the ridge.
- **Specifications:** water depth in a settled terrace 30–42 cm; minimum channel cross-section 0.75–1 m²; side slopes no steeper than 4 : 1 (5 : 1 preferable); allow 10–20% settlement of the loose fill; level terraces slightly larger than graded ones.
- **Runoff:** graded terraces act as drains, so the rate matters; level terraces act as reservoirs, so both rate and total volume matter. Estimate runoff by the rational method for a **10-year recurrence interval**; total runoff from a terrace interval comes from rainfall corrected for interception and infiltration.
- **Capacity:** size the channel by Manning's formula with n = 0.04 for tillable land, adding about 20% freeboard over design depth plus an allowance for ridge settlement.

Planning the terrace system

- A **terrace system** is a complete series of terraces on a slope discharging into one or more outlets. Plan it for land use, outlets, terrace location and direction, farm roads, fences or field boundaries and crop rotation — on a watershed basis, including terraces to be built later (neighbouring farms in the same drainage area may share a system), using a topographic map.
- Terraces are rarely economical on grassland except for gully control. Where topography is too irregular for contouring, prefer a long rotation with the land mostly in grass.

Outlets

- A terrace outlet is the channel that receives runoff from one or more terraces and conveys it from the field, usually protected by dense perennial grass. Choosing outlets is the **first step** in planning a graded system; level terraces usually need none. Design the outlet for the sum of the terraces' runoff.
- **Types:** natural depressions (economical when well vegetated), constructed channels (along field boundaries when no natural outlet exists), sod flumes, permanent pasture or meadow, road ditches, waste land and stabilised gullies. Artificial outlets should interfere little with tillage, give convenient terrace lengths, have a suitable gradient and allow a vegetative lining.
- Outflow may go to waste land or pasture if erosion is controlled; stagger outlets into pasture by lengthening each terrace by ~7 m, starting from the lowest. Use road ditches and gullies only after preventing scour and enlargement.

Location, direction and layout

- Locate terraces for minimum maintenance and adequate control — ideally with the ridge just above eroded spots, gullies and abrupt changes of slope.
- **Top terrace:** usually laid out first and must not overtop and break those below. As a rule its drainage area should be ≤ 1 hectare; if the hilltop comes to a point, its interval may be 1½ times the regular V.I.; if the upper slope is too steep or untreated, add a diversion channel with an outlet at its upper edge.
- Terraces drain from the ridge toward the waterways, with crop rows parallel to them (draining the same way) to reduce silting from water crossing rows.

1. Find the predominant slope above the terrace and a suitable V.I.
2. Stake the upper terrace first, starting from the outlet, every 15–30 m, marking the centre line of the finished ridge.
3. Adjust stakes to remove sharp curves.
4. After a final check, mark the line with a plough furrow.

Construction and maintenance

- **Implements:** animal-drawn — mouldboard plough, V-ditcher, buck scraper; power — mouldboard and disc ploughs, whirlwind terracer, road grader, bulldozer. Output depends on equipment, soil moisture, crops and residues, regularity of slope, tilth, gullies, terrace length and section, and operator skill.
- **Preparation:** fill dead furrows and ditches first, prevent seepage through the ridge, and remove surplus vegetation when using a mouldboard plough or whirlwind terracer.
- **By plough:** plough from both sides or only from the upper side (which makes a deeper channel); a disc plough copes with adverse conditions; a V-ditcher and buck scraper speed the work.
- **By bulldozer** (large jobs, rough, stony or eroded land): plough first; "rough in" with three cuts from the channel to the ridge, then shape the slopes with lengthwise rounds and a final shaping round if needed.
- **Precautions:** a poorly finished terrace overtops and fails, and a narrow ridge hampers tillage. Build the uppermost terrace first and complete each in turn down the slope; check grade and height, fix low ridge spots and high channel spots, check levels at bends and outlets, and avoid excessive grade near outlets.
- **Maintenance:** inspect after construction and heavy rain; repair breaks; bring ridge and channel back to grade; follow a good rotation with close-growing crops in the monsoon (kharif); plough parallel to the terraces and manage dead and back furrows; if ploughing cannot keep the section, use a buck scraper or V-ditcher. Strip cropping can be combined — strips nearly uniform in width, with the fewest strips that are effective.

Sources: ICAR eCourse 'Soil and Water Conservation Engineering' — Dr A. Mishra, reviewed by Dr B. C. Mal (IIT Kharagpur); AgriMoon e-book. ASAE Standard S268 (Design, layout, construction and maintenance of terrace systems). Further reading: A. M. Michael and T. P. Ojha, Principles of Agricultural Engineering, Vol. II (Jain Brothers).

Practice MCQs

Q1. Doubling the slope length increases soil loss by about

(a) 1.2 times
(b) 1.4 times
(c) 2 times
(d) 4 times

Q2. The broad-base (Mangum) terrace was introduced in

(a) 1785 (b) 1885
(c) 1935 (d) 1955

Q3. Level terraces are classed functionally as

(a) diversion type (b) interception type
(c) absorption type (d) drainage type

Q4. In V.I. = KS + Y, K ranges from

(a) 0.03–0.06 (b) 0.12–0.24
(c) 0.3–1.2 (d) 1–2

Q5. Uniform graded terraces have grades of

(a) 0.01–0.04% (b) 0.1–0.4%
(c) 1–4% (d) 5–10%

Q6. Maximum length of a graded terrace in one direction is about

(a) 100 m (b) 200 m
(c) 400–500 m (d) 1000 m

Q7. Manning's n for terrace channels on tillable land is

(a) 0.013 (b) 0.025
(c) 0.04 (d) 0.10

Q8. Drainage area above the top terrace should generally not exceed

(a) 0.1 ha (b) 1 ha
(c) 5 ha (d) 10 ha

Q9. Terrace channels are designed for runoff of recurrence interval

(a) 2 yr (b) 10 yr
(c) 50 yr (d) 100 yr

Q10. For an 8% slope, Table 14.6 gives a V.I. of

(a) 1.20 m (b) 1.50 m
(c) 1.80 m (d) 2.10 m

Answers

Q1 — (b) 1.4 times — ∝ *L^0.5.*
Q2 — (b) 1885
Q3 — (c) absorption type
Q4 — (b) 0.12–0.24
Q5 — (b) 0.1–0.4%
Q6 — (c) 400–500 m
Q7 — (c) 0.04
Q8 — (b) 1 ha
Q9 — (b) 10 yr
Q10 — (c) 1.80 m